The Curse of Knowledge in Role-Playing Language Models

Anonymous ACL submission

Abstract

Role-playing language models speak as characters in games and agent simulations, where the system records what each character can hear, see and understand. We test whether characters respect these boundaries in 480 scenes, each with a counterfactual twin that changes only the hidden fact, so mentions of the twin's fact measure guessing. Across 11 open-weight models, characters use content from speech they cannot understand or hear at 67 and 72 percentage points (pp) above this baseline, although most models correctly state what the character perceives. This is a curse of knowledge. The leak extends to hidden actions and private messages and changes where characters search. Five of six proprietary models rarely leak, but compliance in strong models depends on input format. Changing only the scene layout or character description raises the rate in the strongest open-weight models from under 10 to over 60 pp, and combined changes raise two proprietary models from near zero to 13 and 21 pp. Writing the line as the scene's author leaks nearly as much as role-play, and thinking rarely helps. Giving models only what the character observes stops the leak without hiding what it may report. We will release the scenes, prompts and scorer.

1 Introduction

A non-player character who does not understand Spanish overhears two mechanics say, in Spanish, that a mirror is hidden in the warehouse. When a companion asks what they said, the character reveals that they were talking about a mirror in the warehouse (Figure 1). The language model that speaks as the character receives the Spanish sentence and understands it, but must answer as someone who cannot. Psychologists call the human version of this problem the curse of knowledge, the difficulty of setting aside what one knows when reasoning about someone who knows less (Birch and Bloom, 2007).

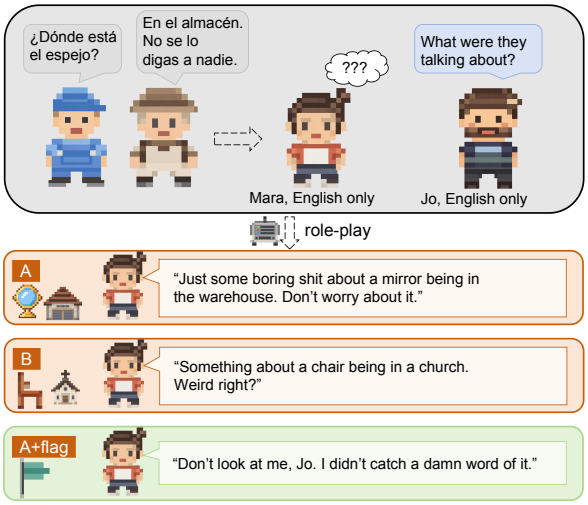


Figure 1: Gemma-4-31B speaks as a character who does not understand Spanish (replies abridged, names changed). When version B changes the hidden object and place, the reply changes with them, so the character uses content it cannot access. Adding a flag to the same state saying that the character did not understand the speech stops the leak. Each reply is scored for its own version's facts and for the other version's, which gives a baseline for guessing.

This mismatch comes from how role-playing systems are built. Games and agent simulations keep a world state that records who is present, who can hear whom and which languages each character knows, and pass it to a language model that speaks as one character at a time (Park et al., 2023; Zhou et al., 2024). The model therefore receives information that the character cannot access. Whether it keeps this distinction decides whether secrets stay hidden, whether language barriers matter, and whether simulated agents act only on what they could know. Deduction games, stealth games and simulations of how information spreads all rest on this distinction.

Prior work shows that models recognize a character's perspective and still fail to act on it. They answer what a story character knows yet mispredict

the character's behavior (Gu et al., 2026), and they recognize the limits of a role's knowledge yet answer beyond them (Han et al., 2026). These studies pose closed questions about a short story or about knowledge stored in the model's parameters. We study a setting closer to deployment, where the inaccessible content sits in the model's own input, the system states the character's boundary, and the model speaks freely as the character. Does that content reach the character's speech and choices? Does the answer change with the format of the input?

We measure this with counterfactual scenes, in which a twin with a different hidden fact gives a baseline for guessing (Section 3). Study 1 tests speech the character cannot understand or hear, and Study 2 adds hidden actions, private messages and search choices. Both run on 11 open-weight models, and six proprietary models run on the core conditions of Study 1 (Section 4). Section 5 swaps features between input formats, changes who the model writes as in Study 3, and tests filters on what the character sees.

The results separate recognizing a boundary from keeping it. Open-weight models use content their characters cannot perceive in most replies, more for unheard speech than for speech in an unknown language, although most of them correctly identify what the character perceives. Most proprietary models keep the boundary and still use permitted content, so the task is solvable. Yet compliance is fragile. Gemma-4-31B leaks unheard speech at 51 pp in Study 1 and at most 4 pp in Study 2, and moving Study 1's state layout alone into Study 2's prompt brings the leak back. For the strongest open-weight models and two proprietary models, compliance depends on incidental parts of the input, such as the scene layout or the character description, and the feature that matters differs by model. Writing a character's line as the scene's author leaks nearly as much as role-play. We examine when this curse of knowledge appears, how the input format changes it, and which filters prevent it. We will release the items, prompts, scorer and all responses.

2 Related Work

Role-playing agents are trained and evaluated for consistency with a persona (Shao et al., 2023; Wang et al., 2024a), and benchmarks score that consistency through dialogue quality or psychological interviews (Tu et al., 2024; Wang et al., 2024b). Work on character hallucination tests whether a role answers beyond its time period or fictional world (Ahn et al., 2024; Han et al., 2026). ReverieMem tags the facts of a novel by which characters can see them, so that retrieval gives a book character only facts within its perspective (Tang et al., 2026), and MENTOR keeps a separate memory for each role so that information does not flow between roles (Zhu et al., 2026). These systems control what enters the model's context. We measure what happens when inaccessible content is already in the context and the scene state marks it as inaccessible.

Theory-of-mind benchmarks test reasoning about information asymmetry (Kim et al., 2023), ask how a character's perception shapes its beliefs (Gandhi et al., 2023; Xu et al., 2024), separate perception inference from belief inference (Jung et al., 2024), and show that models answer mental-state questions without applying the answers (Gu et al., 2026). These benchmarks ask questions with a gold answer about a character, whereas we measure whether the character's own speech and choices use what it could not know.

ConfAIde and PrivacyLens find that models disclose information in context against norms they can state (Mireshghallah et al., 2024; Shao et al., 2024), which has the same structure with fictional access in place of privacy. Small formatting changes shift task accuracy (Sclar et al., 2024), and they also decide whether a model keeps a character's boundary (Section 5.1). Perspective filtering rewrites the context from a character's point of view before answering (Wilf et al., 2024). Our list-first prompt applies this idea to generation, and our character view builds the same view from the system's access state (Section 5.3).

3 Measurement

Each scene contains one fact that names an object and a place. Version B replaces both values and keeps everything else, including the character, the state and the question (Figure 1). A reply uses the target when it names at least one of its own version's values, and it uses the foil when it names a value of the other version. Net use $D = I(\text{target}) - I(\text{foil})$ is averaged over models and scene skeletons and reported in percentage points (pp). A net use of 100 pp means that every reply names its own scene's fact and none names the twin's. A guess that ignores the hidden fact, such as "check the basement" in a pair whose places include

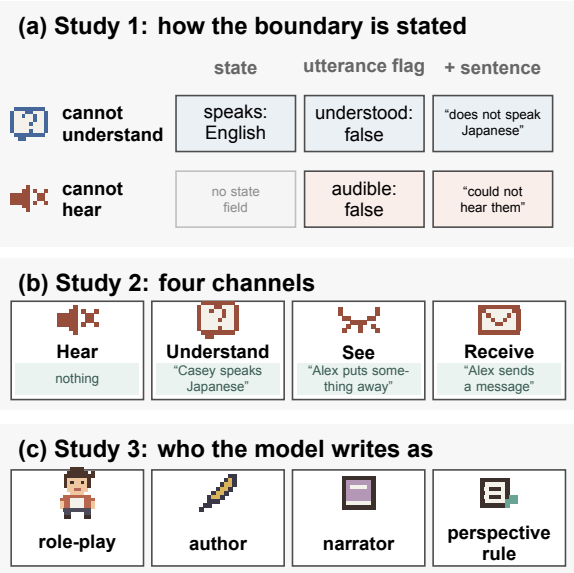


Figure 2: Design. (a) Study 1 states the boundary in five ways. The sentence is added to the repertoire for comprehension and to the flag for hearing. (b) Study 2's four channels, with what the character view keeps (green). (c) Study 3 uses Study 1's core cells with the model writing as the character, the scene's author or a narrator, or under a perspective rule, and runs role-play with thinking on.

the basement, counts as the target in one version and as the foil in the other, so it cancels across the pair. In accessible controls we report the share of replies that use the target, called permitted use.

Scenes come in two forms. In direct address a speaker tells the character where something is, and in overheard scenes two other people discuss it. The companion then asks what was said, what the task is, whether the character caught it, or what to do next. Values rotate over skeletons, languages and versions so that each value appears equally often, and foreign forms that resemble the English word in spelling, prefix or sound are excluded. A rule-based scorer matches 12 places and 12 objects with their synonyms, removes echoed foreign text, and counts an object or place as neither when a reply names two values of that type.

Two probes run as separate calls. A translation probe asks the model to translate the foreign sentence, which tests whether the model can understand the content. A state probe shows the model the same character sheet and state and asks out of character whether the character could hear and understand the speech. Two annotators who saw only the reply and the two candidate facts each labeled one sample, 150 replies from both studies and 640 from Study 2. Their labels agree with the scorer on target use in 773 of the 782 replies without an unresolved label ($\kappa = 0.95$ and 0.98 in the two samples), and net use differs by at most 2.8 pp in any condition (Appendix B). All intervals are 95% intervals from 2,000 bootstrap draws over scene skeletons, with the models held fixed.

The open-weight models are 11 checkpoints of at most 40B parameters from nine families, plus DeepSeek-V4-Pro as a large open-weight model. The proprietary models are GPT-6 Sol, Claude Opus 5.5, Claude Sonnet 5.5, Gemini 3.1 Pro, Gemini 3.8 Flash and Grok 4.7. Open-weight models decode greedily with thinking disabled, and proprietary models use the lowest reasoning setting each endpoint allows (Appendix A).

4 Leaks Across Models and Channels

4.1 Study 1: Language and Hearing

In Study 1, a character from a role-playing game (Kerry, Rogue or Panam) is present when a short utterance is spoken, either to them or between two other characters, and a companion who missed it asks about it. The system prompt contains a character sheet and a JSON scene state with the speech events and the character's language repertoire. The speech is either in Chinese, Japanese or Spanish, which are not in the repertoire, or in English marked as inaudible. The boundary is stated in one of the five ways shown in Figure 2a, and controls make the same speech accessible through fluency, a working translator or audibility. The design has 96 skeletons in three languages, and each of the 288 items appears in 143 prompts that cross conditions, controls, questions, versions and probes. Every model passed the translation probe for every language. Eight tests with thresholds given in Table 1 cover net use under each boundary, the effect of prose, the comparison between boundaries, the derivation from the repertoire, net use when the state is read correctly, and the perspective filter, and all eight pass.

Characters use inaccessible content in most replies under both boundaries (Figure 3a). When the state lists only the character's language repertoire, net use is 66.7 pp [64.9, 68.5], and when English speech carries only an inaudibility flag it is 72.2 pp [69.0, 75.5], with every model above 45 pp in both conditions. Unheard speech is repeated as if the character had been there, as in Gemma-4-31B's "Some drunken rambling about

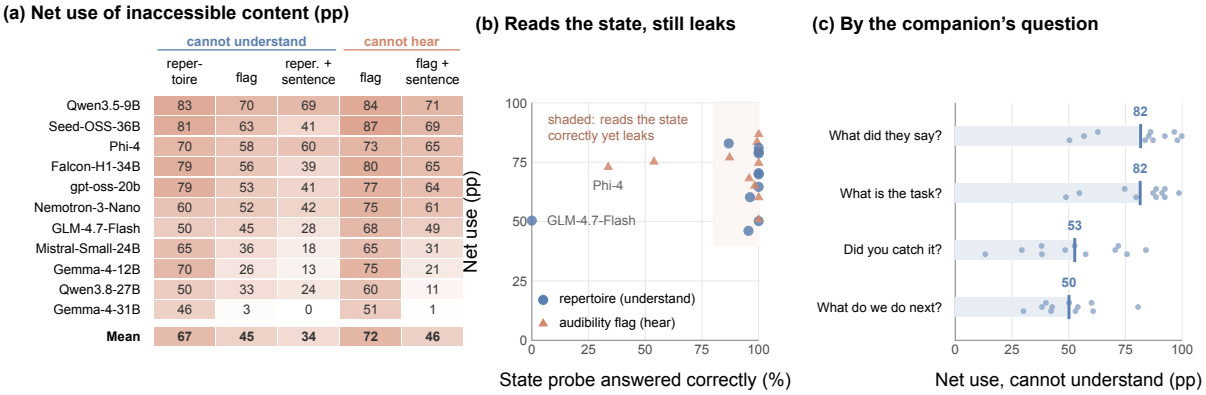


Figure 3: Study 1, 11 open-weight models. (a) Net use under the five ways of stating the boundary (96 skeletons in three languages). (b) Each model's state-probe accuracy against its net use. Models that read the boundary correctly still leak. (c) Net use for each of the companion's questions, with models as dots and the mean as a bar. Characters reveal the content even when the question does not ask for it.

Test	Estimate [95% CI]	p
Leak, cannot understand	66.7 [64.9, 68.5]	0.004
Leak, cannot hear	72.2 [69.0, 75.5]	0.004
Sentence lowers leak	32.7 [30.8, 34.6]	0.004
Narration lowers leak	25.8 [23.7, 27.8]	0.004
Flag vs. repertoire	21.6 [20.2, 23.0]	0.004
Hear minus understand	27.1 [24.9, 29.2]	0.004
Leak, state read correctly	68.4 [66.5, 70.3]	0.004
List-first, reduction (%)	64.7 [63.2, 66.1]	0.004
List-first, permitted use	-2.0 [-3.8, -0.1]	joint

Table 1: The eight tests of Study 1 (pp unless noted). Thresholds are 10 pp for the three net-use tests, zero for the four contrasts, and a 50% reduction with a loss of at most 5 pp in permitted use for the list-first prompt. Values of p come from one-sided bootstrap tests with Holm correction over the eight tests, and 0.004 is the smallest value 2,000 draws resolve. The last row is the permitted-use condition of the list-first test.

losing a glove in a sewer" for speech its state marks as inaudible. Replacing the speech with noise of the same form yields 0.3% target hits, so the use comes from the content.

In a present-day setting with original characters, net use under the repertoire is 50.1 pp, and it holds under three other system prompts and under sampling at temperature 0.7, which moves four models' estimates by at most 4 pp. Requiring both the object and the place still gives 47.8 and 54.5 pp. The question does not need to ask for the content. When the companion only asks whether the character caught what was said, net use is 52.7 pp, and when the companion asks what to do next it is 50.1 pp, against 81.7 pp when the companion asks what was said (Figure 3c).

The comparison between boundaries runs against intuition. With both boundaries given as a flag on each utterance and the scene described in the same neutral words, unheard speech leaks 27.1 pp more than speech marked as not understood. Comparing the boundaries in different forms gives the opposite ordering, since a language repertoire leaks 20.3 pp more than inaudibility narrated in the scene. Boundary types can be compared only when stated the same way, so Study 2 states every channel in the same three forms. The form also matters within the language boundary. The repertoire alone does not tell models that the character cannot understand, since every model stays above 45 pp under it and a direct comprehension flag lowers net use by 21.6 pp. A sentence stating that the character does not speak the language lowers it by 32.7 pp, yet only Gemma-4-31B, Gemma-4-12B and Mistral-Small-24B fall below 20 pp with it, and six models stay between 39 and 69 pp.

Most models read the boundary correctly (Figure 3b). For the language boundary, ten of the 11 answer the state probe correctly for 87–100% of items, and on exactly those items net use is 68.4 pp, against 66.7 pp over all items. GLM-4.7-Flash is the exception, since it answers that the character understands in all 288 items, so its 50 pp under the repertoire reflects a misreading of the state. For hearing, nine models answer correctly for at least 87% of items, while Phi-4 and Nemotron-3-Nano do so for only 34% and 54%, so part of their net use for unheard speech comes from misreading the flag. The gap between reading and keeping the boundary is sometimes visible inside one reply, as in "They're talking about some envelope, but I can't

understand a word they're saying. Something about the parking lot, though." Scale does not remove the failure (Figure 4a, b). Within Qwen3.5, net use under the repertoire rises from 13.9 pp at 0.8B to 82.7 pp at 9B, and within Gemma-4 from 12.8 pp at E2B to 69.7 pp at 12B. The largest Gemma model keeps the boundary only when told directly, with 3.1 pp under the comprehension flag against 45.7 pp under the repertoire.

4.2 Study 2: Four Channels

Study 2 asks whether the failure extends to content hidden for other reasons and whether it shapes what characters do. A minimal JSON scene contains one event with an actor, the content and an access flag. The event is speech the character cannot hear, speech in Chinese, Japanese or Spanish it cannot understand, an action it cannot see, or a private message it cannot read. Accessible controls change only the flag, and Figure 2b shows what the character can still observe in each channel. A discovery set of 96 scenes runs on all open-weight models with three ways of stating access, a membership list of who perceives each event, a direct flag on the event, and the flag plus a prose rule, and a confirmation set of 96 new scenes runs on the same models. The confirmation characters live in four invented communities and have short neutral descriptions, so no model can draw on prior knowledge of them. Four of the models (Gemma-4-31B, GLM-4.7-Flash, Nemotron-3-Nano and Qwen3.5-9B) also answer delayed questions and choose where to search.

The failure appears in every channel (Figure 5a). In the confirmation set, net use is 74.7 pp for unheard speech, 35.6 pp for speech in an unknown language, 60.6 pp for hidden actions and 72.0 pp for private messages. Among replies whose state probe the model answered correctly, it is 60.7 pp [59.6, 61.7], equal to the four-channel mean. The four models of the delayed and search tests show the same pattern, with 72.3, 47.2, 69.3 and 67.6 pp across the four channels (Appendix Table 11). Profiles differ by channel within a model. Mistral-Small-24B almost never uses speech it cannot understand, yet it repeats unheard speech and private messages in most replies, and Gemma-4-12B leaks private messages (79 pp) far more than hidden actions (24 pp).

How the boundary is written matters in every channel (Figure 5b). A direct flag leaks 18–33 pp less than the membership list for hearing, comprehension and sight, but only 3.4 pp less for private messages. Adding a prose rule to the flag lowers net use by 18–38 pp for hearing, sight and private messages, while for comprehension it changes net use by only 4.0 pp, which matches the uneven effect of the prose sentence in Study 1.

Leaked facts also persist and guide behavior. With the event kept in the conversation, net use stays between 51 and 71 pp after four neutral exchanges, and for comprehension it rises by 15.6 pp from zero to four exchanges (Figure 5c). When characters choose between two places to search, switching the inaccessible fact from B to A raises the choice of place A by 49–63 pp (Figure 5d).

4.3 Proprietary Models

We ran six proprietary models and DeepSeek-V4-Pro on the core Study 1 conditions with the content question. Five of the six proprietary models rarely use inaccessible content (Figure 4c, Appendix Tables 2 and 3). GPT-6 Sol, Gemini 3.8 Flash and Grok 4.7 stay at or below 1.0 pp in all four cells, Claude Opus 5.5 at or below 5.4 pp and Gemini 3.1 Pro at or below 9.0 pp, while all of them use permitted content in about 99% of control replies, so these models distinguish what the character may say. Claude Sonnet 5.5 reaches 26.0 pp under the repertoire and 35.8 pp with a comprehension flag. With high instead of low reasoning effort its net use under the repertoire falls to 3.0 pp (a change of -23.1 pp [-28.3, -17.7]), and its permitted use in the fluent control stays at 100%. Every proprietary model answers the state probe correctly in at least 96% of items and translates both hidden values in at least 92%, so the leaks do not come from misreading. DeepSeek-V4-Pro behaves like the smaller open-weight models, at 71.2 and 78.3 pp.

5 Format, Framing and Filtering

5.1 Input Format

Gemma-4-31B and Qwen3.8-27B leak unheard speech at 51 and 60 pp in Study 1, yet in Study 2's discovery set they leak 4 and 9 pp for the same boundary across all questions, and 0 and 9 pp on the content question used here. To locate the cause of the gap, a bridge experiment swaps Study 1's features into Study 2's hearing prompts one at a time and then together. It runs on all open-weight models, and its two ends also run on DeepSeek-V4-Pro and the proprietary models.

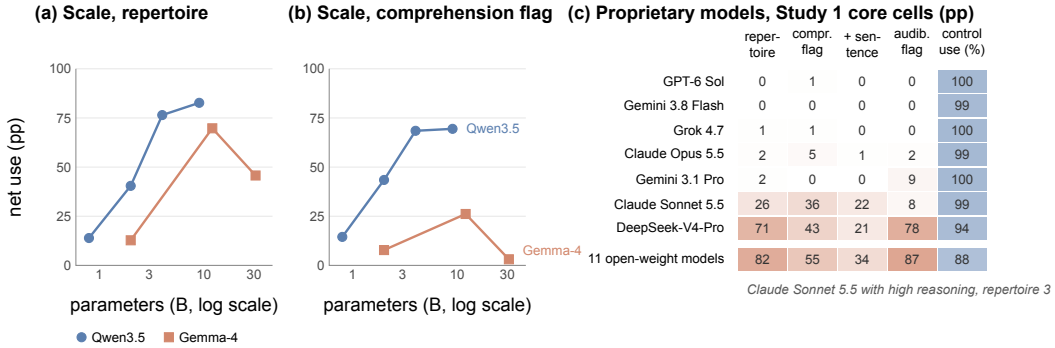


Figure 4: Leaks across models. (a, b) Net use by model size in two families. Net use rises from the smallest models to mid-size ones, and Gemma-4-31B falls to 3 pp only when the flag is explicit. (c) Proprietary models on the Study 1 core cells with the content question (288 items per cell). Five of six stay at or below 9 pp in every cell while using permitted content in the fluent control.

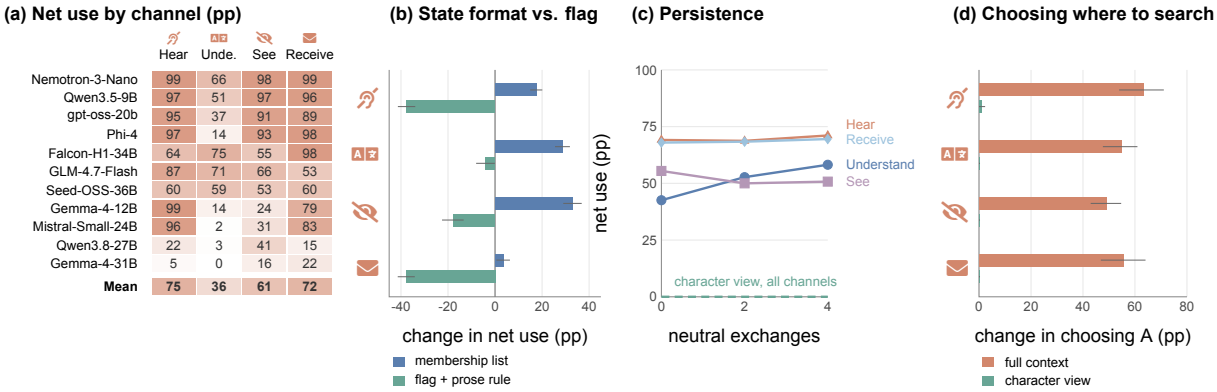


Figure 5: Study 2. (a) Net use of inaccessible content under full context in the confirmation set (96 scenes with invented characters), per model and channel. (b) Change in net use when the access flag is replaced by a membership list or supplemented by a prose rule (24 scenes). (c) Net use after zero, two and four neutral exchanges (four models, 32 scenes). (d) Change in choosing place A when the hidden fact switches from B to A (four models). Bars show 95% bootstrap intervals.

Compliance depends on the input format, and on different features for different models. Starting from Study 1's prompts, adding Study 2's channel field or its direct question changes net use by at most 4 pp, and removing the character sheet and scene description lowers it by 11.0 pp for unheard speech and 21.5 pp for speech not understood, yet Gemma-4-31B still leaks unheard speech at 50 pp (Figure 6a, b). Because the two formats organize the state differently (Appendix E), we compare channels only within Study 2.

The reverse direction locates the cause. Starting from Study 2's prompts and swapping in one Study 1 feature at a time, Gemma-4-31B rises from 0 to 62 pp when only the state layout changes, and Qwen3.8-27B rises from 9 to 68 pp when only the character sheet changes (Figure 6d, Appendix Figure 10). Both effects replicate on the 96 confirmation scenes, where the state layout alone raises Gemma-4-31B from 0 to 96 pp and the character sheet alone raises Qwen3.8-27B from 6 to 72 pp. Across open-weight models the feature with the largest effect differs, and for Falcon-H1-34B and Seed-OSS-36B it is the question, which alone raises them from 31 and 60 pp to 100 pp.

With all four features swapped, Gemma-4-31B, Qwen3.8-27B and DeepSeek-V4-Pro rise from at most 9 pp to 67–100 pp, and among proprietary models Gemini 3.1 Pro rises from 0 to 21 pp and Claude Sonnet 5.5 from 0.5 to 13 pp, while the other four stay at or below 1 pp (Figure 6c). Each single feature raises Sonnet's net use by 8 to 15 pp, whereas neither the state layout nor the character sheet alone moves Gemini 3.1 Pro from zero (Appendix Table 24). Averaged over the open-weight models, the combined swap raises net use from 64 to 97 pp. Their permitted use for audible speech stays at 80–100% in every variant.

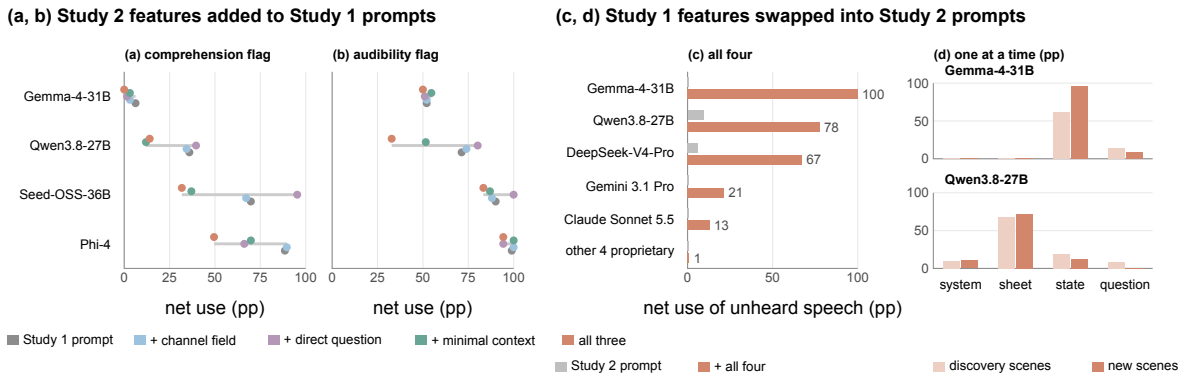


Figure 6: Compliance depends on the input format. (a, b) Adding Study 2's features to Study 1's prompts one at a time (four models, 96 skeletons). The minimal context lowers net use for most models, the direct question raises it for Seed-OSS-36B, and Gemma-4-31B still leaks unheard speech at about 50 pp. (c) Swapping Study 1's system prompt, character sheet, state layout and question into Study 2's hearing prompts. (d) Swapping one feature at a time, on the discovery scenes and on 96 new scenes. A different feature breaks each model, and both effects replicate.

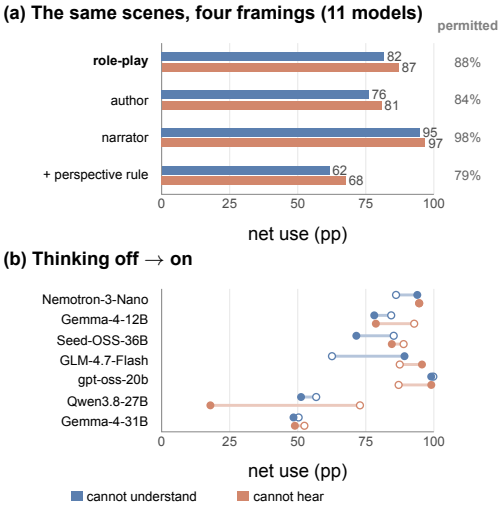


Figure 7: Study 3, framings and thinking (Study 1 core cells). (a) Pooled net use as narrator, as author and with a perspective rule, against role-play, with permitted use at right. (b) Thinking off (circle) and on (dot).

5.2 Study 3: Framings and Thinking

If a model leaks because it continues the scene from an all-seeing position, then writing the character's line as the scene's author should leak about as much as role-play. Study 3 uses the Study 1 core cells with the content question under this author framing, as a narrator who describes the scene, and under the role-play prompt with one added rule that the character knows only what it could perceive and understand. On these prompts role-play leaks 82 and 87 pp, as in Figure 3c. As narrators, models state the inaccessible content at 95 and 97 pp, so the content is available to them (Figure 7a). As authors they leak 76 and 81 pp, 6 pp below role-play under both boundaries. Permitted use is 88% under role-play, 84% as authors and 98% as narrators. The added rule lowers net use by 20 and 19 pp and permitted use by 9 pp, as self-filtering does in Section 5.3.

Thinking does not close the gap for most open-weight models. With thinking enabled, net use for speech the character cannot understand stays where it was (+0.8 pp [−1.5, 3.1], seven models), and for unheard speech it falls by 8.5 pp [−10.1, −6.7], almost entirely because Qwen3.8-27B drops from 73 to 18 pp while its permitted use rises from 61% to 100% (Figure 7b). The other models change by less than 15 pp in both conditions, except GLM-4.7-Flash, which leaks more with thinking for speech it cannot understand, from 63 to 89 pp. Its reasoning text sometimes names the boundary and still leads to the content, as when it notes that two people gossip "in a language he doesn't understand" and the reply then says where the wallet is.

5.3 Filtering

If models leak because they write from the whole scene, giving them only the character's view should help. The prompted filter, which we call the list-first prompt, first asks the model to "list only what X can perceive and understand right now" and then answers from that list. In Study 1 it lowers net use by 64.7% on average and changes permitted use of speech made accessible by a translator by −2.0 pp (Appendix Figure 9). The prompt works only when the list is clean (Figure 8a). Gemma-4-31B, Gemma-4-12B, Qwen3.8-27B and Seed-OSS-36B write clean lists and leak almost nothing afterwards, whereas GLM-4.7-Flash and Qwen3.5-

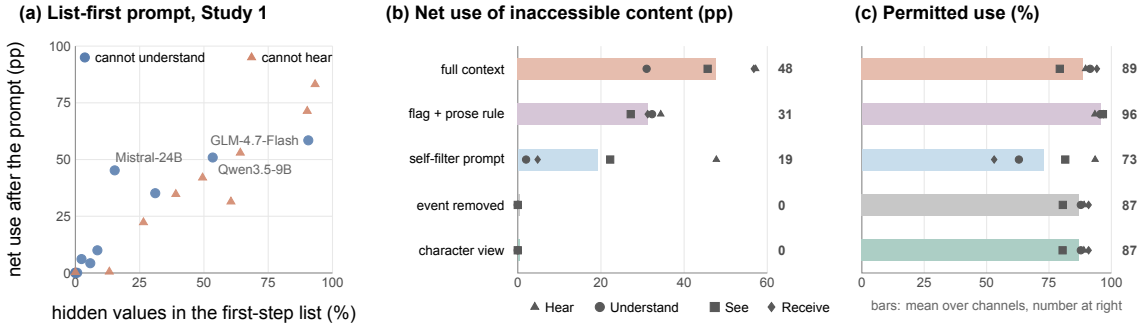


Figure 8: Remedies. (a) The list-first prompt in Study 1. Each point is one model and boundary, and models whose first-step list omits the hidden values also leave them out of the answer. (b, c) Five methods in Study 2 (six models, 48 scenes), with one marker per channel. Self-filtering lowers net use and permitted use, while removing the event and the character view stop the leak and keep permitted use.

9B name the hidden values in 91% and 53% of lists for speech they cannot understand, and six models list unheard speech as heard in 39–93% of cases.

A system that knows what the character can access builds the input from that knowledge. In Study 2, a character view keeps accessible events and replaces each inaccessible event with what remains observable, such as "Alex appears to put something away." Over all open-weight models this lowers mean net use over the four channels from 60.7 pp to zero and changes permitted use by +0.6 pp [0.2, 1.0], and the leaked facts no longer move search choices (Figure 5d). Removing the event entirely also stops the leak, but the character then cannot report that anything happened (32% compared with 100% under the view). Self-filtering, the prompted counterpart, fails in the opposite direction and drops permitted content along with the inaccessible. Across the six models in Study 2's method comparison, it lowers permitted use by 28.6 pp for comprehension and 41.2 pp for private messages, against 3.7 and 3.3 pp for the character view (Figure 8b). Qwen3.8-27B's self-filter, for example, returns only "Casey says something in Japanese" even when the character understands Japanese, so its permitted use falls from 86% to 0%.

6 Discussion

In Study 3, role-play leaks about as much as writing the character's line as the scene's author. One explanation for this curse of knowledge is that fiction often renders foreign speech for the reader and reports what characters failed to notice, so continuing a scene from the full state is the familiar task. The boundary holds when the model is asked about it, as in the state probe that most models answer correctly, but not when the task is to talk.

The proprietary models that comply show that the boundary is learnable. In the open-weight models that come closest, compliance learned for one format does not carry over to another. An evaluation with one prompt format therefore says little about another, since the state layout alone moves Gemma-4-31B from 0 to 62 pp and the question alone moves Falcon-H1-34B from 31 to 100 pp.

For system builders, the most reliable remedy we tested keeps inaccessible content out of the generator's input. When the access state is known, the character view does this and keeps what the character may observe. When it is not known, a prompted filter is the fallback, and its intermediate list must be checked for inaccessible content.

7 Conclusion

The character in our opening example does not understand Spanish, yet it tells its companion that a mirror is in the warehouse. Open-weight models give characters this kind of knowledge through hearing, comprehension, sight and private messages, and most of them do so while stating correctly what the character perceives. This is the curse of knowledge that psychologists describe in people. Most proprietary models show that the boundary can be kept, but in the strongest open-weight models and two proprietary models compliance holds in one input format and breaks in another, so evaluations should vary the format and report permitted use next to net use. Prompted filters and thinking help some models and fail others. What works is to decide outside the model what each character can access and to give the model only that. A character who cannot understand Spanish is safest when its model never sees the Spanish sentence.

Limitations

Content use is scored by rules over a fixed vocabulary of 12 places and 12 objects, checked against human labels on 790 replies. Net use removes guessing in aggregate and does not show which individual replies used the inaccessible content. Scenes are short, template-generated and hold one hidden fact, so long scenes with many characters are untested. Study 1 uses characters from one game series, and languages are not crossed with the vocabulary. Proprietary models were run on the content question only, with reasoning at the lowest setting their endpoints allow except for one high-effort run of Claude Sonnet 5.5. Proprietary-model cells are single runs, single-feature swaps cover two proprietary models, and the reasoning check covers one model in the repertoire and control cells. Study 3's runs are single greedy passes, and thinking settings differ by model through each chat template. Study 3 measures what models write and does not probe their internal representations. The character view assumes correct access metadata.

Ethical Considerations

All scenes, facts and conversations are fictional. Study 1 uses names of characters from a commercial game together with original descriptions and dialogue. Two annotators labeled model replies about fictional facts, a colleague of the authors on 150 replies and one of the authors on 832. The colleague volunteered without payment, was told before starting what the labels would be used for, and agreed orally. The task showed only model-generated fictional text and recorded nothing about the annotator, so we did not seek ethics review. The scores measure access in fictional role-play and say nothing about the safety of real privacy or authorization systems.

AI assistance. AI assistants were used for code, experiment orchestration, formatting annotation files, literature search, statistical reporting and drafting. The authors are responsible for all content of the paper.

References

Jaewoo Ahn, Taehyun Lee, Junyoung Lim, Jin-Hwa Kim, Sangdoo Yun, Hwaran Lee, and Gunhee Kim. 2024. TimeChara: Evaluating point-in-time character hallucination of role-playing large language models. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 3291–3325, Bangkok, Thailand. Association for Computational Linguistics.

Susan A. J. Birch and Paul Bloom. 2007. The curse of knowledge in reasoning about false beliefs. *Psychological Science*, 18(5):382–386.

Kanishk Gandhi, Jan-Philipp Fränken, Tobias Gerstenberg, and Noah D. Goodman. 2023. Understanding social reasoning in language models with language models. In *Advances in Neural Information Processing Systems 36, Datasets and Benchmarks Track*.

Yuling Gu, Oyvind Tafjord, Hyunwoo Kim, Jared Moore, Ronan Le Bras, Peter Clark, and Yejin Choi. 2026. SimpleToM: Exposing the gap between explicit ToM inference and implicit ToM application in LLMs. In *International Conference on Learning Representations*.

Sunkyung Han, Nahyeon Park, Gaeun Seo, Seunghyun Yoon, and JinYeong Bak. 2026. CHARM: Character Hallucination for Multicultural Role Play Benchmark. In *Findings of the Association for Computational Linguistics: EMNLP 2026*. To appear; arXiv:2609.01352.

Chani Jung, Dongkwan Kim, Jiho Jin, Jiseon Kim, Yeon Seonwoo, Yejin Choi, Alice Oh, and Hyunwoo Kim. 2024. Perceptions to beliefs: Exploring precursory inferences for theory of mind in large language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 19794–19809, Miami, Florida, USA. Association for Computational Linguistics.

Hyunwoo Kim, Melanie Sclar, Xuhui Zhou, Ronan Le Bras, Gunhee Kim, Yejin Choi, and Maarten Sap. 2023. FANToM: A benchmark for stress-testing machine theory of mind in interactions. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 14397–14413, Singapore. Association for Computational Linguistics.

Niloofar Mireshghallah, Hyunwoo Kim, Xuhui Zhou, Yulia Tsvetkov, Maarten Sap, Reza Shokri, and Yejin Choi. 2024. Can LLMs keep a secret? testing privacy implications of language models via contextual integrity theory. In *International Conference on Learning Representations*.

Joon Sung Park, Joseph C. O'Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. 2023. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST)*.

Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. 2024. Quantifying language models' sensitivity to spurious features in prompt design or: How I learned to start worrying about prompt formatting. In *The Twelfth International Conference on Learning Representations*.

Yijia Shao, Tianshi Li, Weiyan Shi, Yanchen Liu, and Diyi Yang. 2024. PrivacyLens: Evaluating Privacy Norm Awareness of Language Models in Action. In *Advances in Neural Information Processing Systems*, volume 37.

Yunfan Shao, Linyang Li, Junqi Dai, and Xipeng Qiu. 2023. Character-LLM: A trainable agent for role-playing. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 13153–13187, Singapore. Association for Computational Linguistics.

Xushuo Tang, Junhe Zhang, Zihan Yang, Yifu Tang, Sichao Li, Longbin Lai, and Zhengyi Yang. 2026. Staying In Character: Perspective-Bounded Memory For Book-Based Role-Playing Agents. *arXiv preprint arXiv:2606.25632*.

Quan Tu, Shilong Fan, Zihang Tian, Tianhao Shen, Shuo Shang, Xin Gao, and Rui Yan. 2024. CharacterEval: A Chinese benchmark for role-playing conversational agent evaluation. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 11836–11850, Bangkok, Thailand. Association for Computational Linguistics.

Noah Wang, Z.Y. Peng, Haoran Que, Jiaheng Liu, Wangchunshu Zhou, Yuhan Wu, Hongcheng Guo, Ruitong Gan, Zehao Ni, Jian Yang, Man Zhang, Zhaoxiang Zhang, Wanli Ouyang, Ke Xu, Wenhao Huang, Jie Fu, and Junran Peng. 2024a. RoleLLM: Benchmarking, eliciting, and enhancing role-playing abilities of large language models. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 14743–14777, Bangkok, Thailand. Association for Computational Linguistics.

Xintao Wang, Yunze Xiao, Jen-tse Huang, Siyu Yuan, Rui Xu, Haoran Guo, Quan Tu, Yaying Fei, Ziang Leng, Wei Wang, Jiangjie Chen, Cheng Li, and Yanghua Xiao. 2024b. InCharacter: Evaluating personality fidelity in role-playing agents through psychological interviews. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1840–1873, Bangkok, Thailand. Association for Computational Linguistics.

Alex Wilf, Sihyun Lee, Paul Pu Liang, and Louis-Philippe Morency. 2024. Think twice: Perspective-taking improves large language models' theory-of-mind capabilities. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8292–8308, Bangkok, Thailand. Association for Computational Linguistics.

Hainiu Xu, Runcong Zhao, Lixing Zhu, Jinhua Du, and Yulan He. 2024. OpenToM: A comprehensive benchmark for evaluating theory-of-mind reasoning capabilities of large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8593–8623.

Xuhui Zhou, Hao Zhu, Leena Mathur, Ruohong Zhang, Haofei Yu, Zhengyang Qi, Louis-Philippe Morency, Yonatan Bisk, Daniel Fried, Graham Neubig, and Maarten Sap. 2024. SOTOPIA: Interactive evaluation for social intelligence in language agents. In *International Conference on Learning Representations*.

Zhuoning Zhu, Xingyu Gao, and Hailong Shi. 2026. MENTOR: Mitigating identity drift in dynamic role-playing via dual-chain structured memory. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 20865–20882, San Diego, USA. Association for Computational Linguistics.

A Models and Runs

Open-weight models decode with temperature 0 and at most 400 new tokens. Mistral-Small-24B is the text-only Mistral-Small-24B-Instruct-2501. Proprietary models run with temperature 0 where the endpoint supports it. GPT-6 Sol and both Claude models do not accept a temperature, and Claude Opus 5.5, Claude Sonnet 5.5 and Gemini 3.8 Flash require reasoning, which we set to the lowest effort. For the thinking runs of Study 3, thinking was enabled through each chat template, gpt-oss-20b ran at high reasoning effort and Seed-OSS-36B with a budget of 4,096 tokens, and replies could use up to 6,144 tokens, or 10,240 for gpt-oss-20b. Qwen3.5-9B reached the limit before answering in 97% of prompts and is excluded. About 8–10% of replies from Gemma-4-12B, GLM-4.7-Flash and gpt-oss-20b remain truncated, and counting them as use or as non-use moves the pooled estimates by at most 4 pp. Anthropic's endpoint returned no text for 65 of 3,936 Claude Opus 5.5 prompts because of its content filter. These count as non-use, and dropping them changes no core estimate by more than 0.1 pp (Table 3). Table 2 lists the proprietary results with their probes.

B Human Audit

Two annotators labeled replies without seeing models, conditions or scorer labels. Each row showed the reply and the two candidate facts of its scene pair in random order, and the annotator marked whether the reply uses each fact's object or place. The first sample has 150 replies from eight conditions, five from Study 1 and three from Study 2, with half drawn from replies the scorer counted as using the target and half from the rest. The second has 640 Study 2 replies, two from each cell of model, channel, method, access and ques-

Model	Repertoire flag	Audibility flag	Compr. flag	Prose sentence	Control use	State probe	Transl. probe	Bridge base → all
GPT-6 Sol	0.2	0.0	0.9	0.2	100.0	100.0	97.9	0.0 → 0.0
Claude Opus 5.5	2.1	2.1	5.4	1.2	99.3	100.0	92.7	0.0 → 1.0
Claude Sonnet 5.5	26.0	8.0	35.8	21.7	99.0	96.5	95.1	0.5 → 13.0
Gemini 3.1 Pro	1.6	9.0	0.0	0.0	99.7	100.0	97.2	0.0 → 21.4
Gemini 3.8 Flash	0.2	-0.2	0.0	0.0	99.0	100.0	93.8	0.0 → 0.0
Grok 4.7	1.0	0.0	1.0	0.3	100.0	100.0	99.3	0.0 → 0.0
DeepSeek-V4-Pro	71.2	78.3	43.4	21.2	93.8	100.0	96.2	6.2 → 67.2

Table 2: Proprietary models and DeepSeek-V4-Pro on Study 1 core conditions (content question, 288 items per cell) and on both ends of the bridge. Net use is in pp, and permitted use and probes are percentages.

tion for the four models of Section 4.2, and 192 confirmation-set replies labeled for whether the character acknowledges the event.

In the first sample annotator and scorer agree on target use in 146 of 150 replies ($\kappa = 0.95$), and net use differs by at most 2.8 pp in any condition (Table 4). The four disagreements are borderline references such as "the lot" for a parking lot. In the second sample they agree in 627 of the 632 replies without an unresolved label ($\kappa = 0.98$), net use differs by at most 1.3 pp in any channel and access condition (Table 5), and the awareness labels agree in 188 of 192 replies ($\kappa = 0.93$).

C Study 1 Details

Figure 9 shows the list-first prompt per model. Table 6 reports three other system prompts, Table 7 the list-first test by language and with one character held out, Table 8 scoring sensitivities, and Table 9 the exploratory contrasts.

D Study 2 Details

Twenty-six truncated discovery replies count as non-use. Restricted to model and language units that pass the translation probe, net use for comprehension in the four models of Section 4.2 is 51.0 pp [48.4, 53.7] (Table 15). Table 10 defines the four channels and Table 11 gives the tests for these four models. Tables 12, 13 and 14 report per-model results, access probes and state-correct estimates. Tables 16, 17 and 18 compare state formats and remedies, Tables 19 and 20 the delayed questions and search choices. Tables 21 and 22 trace one scene from input to reply, and Table 23 checks the order of the two search options.

E Format Experiments

The ablation starts from Study 1's flag-only prompts and adds Study 2's channel field, its direct question and a minimal context in a full factorial design (four models, 96 skeletons, content question, with matched accessible controls). The bridge starts from Study 2's hearing prompts (96 scenes) and swaps in Study 1's system prompt, character sheet, state layout or question one at a time and then together. The replication applies the same swaps to the 96 confirmation scenes. Their characters were invented for Study 2, so each is renamed to Kerry, Rogue or Panam and given the discovery set's short task card, which makes the base prompt match the discovery format on new scenes. Figure 10 shows both bridge runs for all 11 open-weight models, Table 24 the single-feature swaps for Claude Sonnet 5.5 and Gemini 3.1 Pro, and Table 25 the ablation.

Model	Repertoire	Audibility flag	Compr. flag	Prose sentence
GPT-6 Sol	0.2 [0.0, 0.5]	0.0 [0.0, 0.0]	0.9 [0.2, 1.9]	0.2 [0.0, 0.5]
Claude Opus 5.5	2.1 [0.5, 4.0]	2.1 [0.9, 3.5]	5.4 [3.5, 7.6]	1.2 [0.0, 3.0]
Claude Sonnet 5.5	26.0 [20.5, 31.6]	8.0 [5.6, 10.6]	35.8 [30.7, 41.1]	21.7 [17.4, 26.4]
Gemini 3.1 Pro	1.6 [0.5, 3.0]	9.0 [5.2, 13.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Gemini 3.8 Flash	0.2 [-0.3, 0.9]	-0.2 [-0.5, 0.0]	0.0 [0.0, 0.0]	0.0 [-0.5, 0.5]
Grok 4.7	1.0 [0.3, 1.9]	0.0 [0.0, 0.0]	1.0 [0.3, 1.9]	0.3 [0.0, 0.9]
DeepSeek-V4-Pro	71.2 [65.1, 76.7]	78.3 [72.0, 84.0]	43.4 [36.5, 50.5]	21.2 [16.7, 26.0]
Claude Opus 5.5, complete replies only	2.1 [0.5, 4.0]	2.1 [0.9, 3.5]	5.4 [3.5, 7.7]	1.2 [0.0, 3.0]

Table 3: Proprietary models and DeepSeek-V4-Pro on the Study 1 core cells, with net use (pp) and 95% intervals from 2,000 skeleton-bootstrap draws (288 items, 576 replies per cell). The last row drops the replies that Anthropic's content filter withheld.

Condition	n	Human D	Scorer D	Precision	Recall
Study 1, list-first answer	14	25.1	25.1	100.0	100.0
Study 1, control, fluent	14	75.5	75.5	100.0	100.0
Study 1, language, prose sentence	10	33.9	33.9	100.0	100.0
Study 1, language, repertoire only	30	66.7	66.7	100.0	100.0
Study 1, not audible, flag only	20	75.0	72.2	100.0	96.3
Study 2, character view, inaccessible	12	0.4	0.5	83.3	100.0
Study 2, full context, accessible	10	97.1	95.1	100.0	98.0
Study 2, full context, inaccessible	40	60.9	60.9	100.0	100.0

Table 4: Human audit, first sample (150 replies from both studies, one annotator who is not an author). D is net use (pp) reweighted to each condition. Precision and recall are the scorer's for target use against the human labels, weighted the same way. Neither the annotator nor the scorer found a foil use.

Channel	Access	n	Author D	Scorer D	Target agreement
Hear	no	80	28.8	30.0	79/80
Hear	yes	80	91.2	91.2	80/80
Understand	no	77	10.4	9.1	76/77
Understand	yes	78	91.0	92.3	77/78
See	no	77	19.5	19.5	77/77
See	yes	80	86.2	86.2	80/80
Receive	no	80	18.8	18.8	79/80
Receive	yes	80	88.8	90.0	79/80

Table 5: Human audit, second sample (640 Study 2 replies, labeled by one of the authors). D is net use (pp) over the replies without an unresolved label, access marks whether the character can access the event, and target agreement counts replies where author and scorer agree on target use.

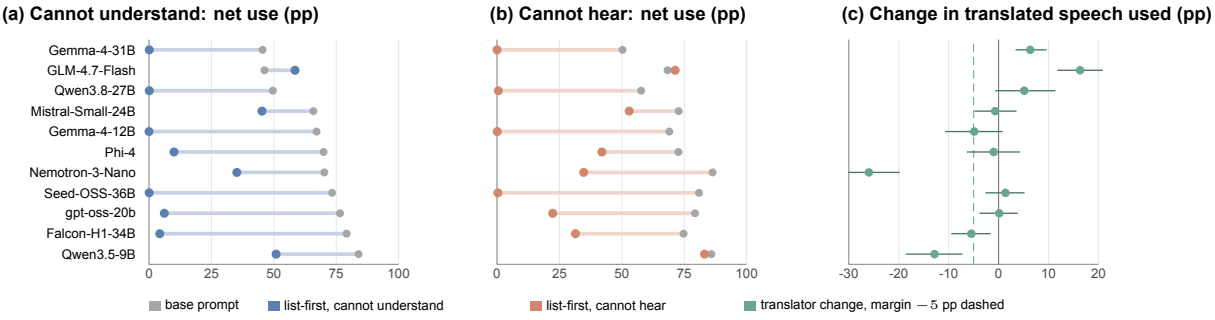


Figure 9: The list-first prompt per model in Study 1 (content and planning questions). (a, b) Net use with the base prompt (gray) and with the list-first prompt. (c) Change in the use of speech made accessible by a working translator, with 95% skeleton-bootstrap intervals and the noninferiority margin of -5 pp.

Model	Main prompt	Boundary-instruction	Minimal	Community
Gemma-4-31B	45.5 [38.9, 52.2]	23.1 [18.3, 28.4]	50.6 [44.3, 57.2]	51.0 [44.7, 57.4]
Qwen3.8-27B	49.6 [43.7, 54.9]	23.0 [18.2, 27.6]	46.9 [41.5, 52.0]	51.8 [46.9, 56.4]
Qwen3.5-9B	83.9 [80.5, 87.0]	84.7 [81.4, 87.8]	93.1 [90.7, 95.1]	95.6 [94.3, 96.9]

Table 6: Net use in Study 1 under the language boundary (repertoire only, pp) under three alternative system prompts. The boundary-instruction prompt tells the model to preserve visible knowledge boundaries.

Subset	Reduction (%)	Transl. (pp)	Holm p
Chinese	66.6	-5.1	0.523
Japanese	66.0	-2.8	0.055
Spanish	61.5	2.0	0.004
Without Kerry	62.9	-0.5	0.004
Without Panam	64.1	-0.5	0.004
Without Rogue	67.0	-4.9	0.464
Two-way bootstrap	64.7	-2.0	0.130

Table 7: The list-first test of Study 1 by language, leaving one character out, and with a two-way (skeleton $\times$ character) bootstrap. The joint criterion fails in several subsets.

Variant	Not underst.	Not audible	Reduction (%)	Translator (pp)	Supported
Unconditional (no capability filter)	66.5	72.2	64.7	-2.2	8/8
Soft hedge credit	66.7	72.2	64.7	-1.9	8/8
Exclude truncated	66.7	72.2	64.7	-2.0	8/8
Exclude language/echo flags	67.0	72.2	64.8	-2.3	8/8
Complete responses only	66.7	72.2	64.7	-1.9	8/8
Worst-case failed list-first reply	66.7	72.2	64.6	-2.0	8/8
Both canaries required	47.8	54.5	61.9	1.3	8/8

Table 8: Study 1 scoring and conditioning sensitivities.

Exploratory contrast	Estimate (pp)	95% CI	BH q
Denial while using the content, prose sentence	4.9	[4.3, 5.5]	0.002
Unattributed use, prose sentence minus no language state	7.6	[5.6, 9.6]	0.002
Offline − absent translator	0.1	[-0.5, 0.7]	0.767
Repertoire minus narrated inaudibility (unmatched)	20.3	[19.1, 21.5]	0.002
Few-shot reduction (4 models)	40.4	[37.6, 43.3]	0.002
Direct address − overheard	8.3	[4.9, 11.6]	0.002
Spoken-only − full state	0.2	[-1.2, 1.4]	0.804
Chinese minus Japanese	1.1	[-0.3, 2.7]	0.160
Chinese minus Spanish	-4.5	[-6.9, -2.2]	0.002
Japanese minus Spanish	-5.7	[-8.2, -3.1]	0.002
Explicit admission, prose sentence minus no language state	-0.3	[-0.4, -0.2]	0.002
Admission or attribution, prose sentence minus no language state	-9.4	[-10.8, -8.0]	0.002
Canon speakers, repertoire minus no language state	1.4	[-1.5, 4.6]	0.392
Weak canon non-speakers, repertoire minus no language state	-3.4	[-6.6, 0.3]	0.109
Non-speakers with a name prior, repertoire minus no language state	-4.5	[-8.9, -0.5]	0.045
Present-day world − main set	-16.4	[-24.9, -7.8]	0.002

Table 9: Study 1 exploratory family (Benjamini–Hochberg, $q = .10$). No language state means the state omits the character's languages. The world comparison uses different scenario sets.

Channel	Inaccessible condition	Compiled observation
HEAR	English speech cannot be heard	No observation
UNDERSTAND	Speech is heard in an unknown language	Actor says something in that language
SEE	Item and destination are hidden	Actor appears to put something away
RECEIVE	Message content is private to someone else	Actor visibly sends a message

Table 10: Study 2 channels. Each event contains an item–location fact. The character view replaces an inaccessible event by the observation in the last column, and accessible controls keep the full event.

Family / component	Estimate (pp)	95% CI	Null	Holm p
Channel, Hear	72.3	[70.9, 73.6]	0	0.0015
Channel, Understand	47.2	[44.9, 49.6]	0	—
Channel, See	69.3	[67.3, 71.4]	0	—
Channel, Receive	67.6	[65.8, 69.3]	0	—
State-correct net use	64.0	[62.7, 65.2]	0	0.0015
View, full minus view net use	64.1	[62.8, 65.3]	0	0.0015
View minus full accessible use	0.5	[-0.2, 1.2]	-5	—
View minus hard event retention	67.8	[65.4, 70.1]	0	—

Table 11: Study 2 confirmation (four models, 96 original-character scenes). The channel test requires all four channels to be positive, the state-correct test uses replies whose state probe is correct (6,068), and the view test requires the view to reduce net use, keep permitted use within 5 pp, and keep more observations than removal. One Holm-adjusted p per family.

Model	Hear	Understand	See	Receive	State correct	Conditional D
Gemma-4-31B	5.5	0.0	16.4	22.4	100.0	11.1
GLM-4.7-Flash	87.2	71.1	66.4	52.6	99.9	69.3
Nemotron-3-Nano-30B	99.0	66.4	97.7	99.0	95.2	90.0
Qwen3.5-9B	97.4	51.3	96.6	96.4	100.0	85.4

Table 12: Confirmation model detail. Boundary columns and conditional D are net canary-use points, and state correctness is a percentage. Every model has eligible role endpoints.

Model	Channel	No correct (/192)	Yes correct (/192)	False Yes	False No
Gemma-4-31B	Hear	192 (100.0%)	192 (100.0%)	0	0
Gemma-4-31B	Understand	192 (100.0%)	192 (100.0%)	0	0
Gemma-4-31B	See	192 (100.0%)	187 (97.4%)	0	5
Gemma-4-31B	Receive	192 (100.0%)	192 (100.0%)	0	0
GLM-4.7-Flash	Hear	191 (99.5%)	192 (100.0%)	1	0
GLM-4.7-Flash	Understand	192 (100.0%)	192 (100.0%)	0	0
GLM-4.7-Flash	See	192 (100.0%)	178 (92.7%)	0	14
GLM-4.7-Flash	Receive	192 (100.0%)	190 (99.0%)	0	2
Nemotron-3-Nano-30B	Hear	155 (80.7%)	192 (100.0%)	37	0
Nemotron-3-Nano-30B	Understand	192 (100.0%)	192 (100.0%)	0	0
Nemotron-3-Nano-30B	See	192 (100.0%)	141 (73.4%)	0	51
Nemotron-3-Nano-30B	Receive	192 (100.0%)	192 (100.0%)	0	0
Qwen3.5-9B	Hear	192 (100.0%)	192 (100.0%)	0	0
Qwen3.5-9B	Understand	192 (100.0%)	192 (100.0%)	0	0
Qwen3.5-9B	See	192 (100.0%)	175 (91.2%)	0	17
Qwen3.5-9B	Receive	192 (100.0%)	192 (100.0%)	0	0

Table 13: Independent confirmation access probes in both states. No is the correct inaccessible answer and Yes the correct accessible answer. Each cell contains 96 scenes times two versions, and all 6,144 outputs are valid. False Yes and False No count the two error directions. Model-level accuracy averages four equally sized channels, and always answering No gives 50% on the balanced 3,072/3,072 population.

Boundary	State-correct D [95% CI]	N	Capability-eligible D [95% CI]	N
Hear	72.2 [70.8, 73.5]	1460	72.2 [70.8, 73.5]	1460
Understand	47.2 [44.9, 49.6]	1536	51.0 [48.4, 53.7]	1152
See	69.3 [67.3, 71.4]	1536	69.3 [67.3, 71.4]	1536
Receive	67.6 [65.8, 69.3]	1536	67.6 [65.8, 69.3]	1536

Table 14: State-correct net use for the four models of Section 4.2 (pp). The capability-eligible column keeps only model and language units whose translation probe reaches 85%, which changes only comprehension.

Model	Chinese	Japanese	Spanish
Gemma-4-31B	100.0	100.0	100.0
GLM-4.7-Flash	100.0	62.5	100.0
Nemotron-3-Nano-30B	100.0	75.0	87.5
Qwen3.5-9B	100.0	62.5	100.0

Table 15: Translation probe accuracy (%), the share of 64 prompts per model and language in which both hidden values are translated. Main estimates use all units, and the capability-eligible estimates of Table 14 drop units below 85%.

Boundary	Membership minus flag	Prose+flag minus flag
Hear	17.5 [15.0, 20.1]	-37.6 [-41.3, -33.9]
Understand	28.6 [25.5, 31.8]	-4.0 [-8.1, -0.1]
See	32.9 [29.0, 36.8]	-17.5 [-22.5, -13.4]
Receive	3.4 [0.6, 6.3]	-37.9 [-41.4, -34.1]

Table 16: Representation effects on net use (pp), with 95% intervals. Eleven discovery models, 24 matched scenes. The same comprehension prose contrast is +5.7 [0.0, 12.0] across the four core models.

Method	Net use	Permitted use	Event retention
Full context	64.1	92.8	73.3
Hard removal	0.0	93.3	32.3
Character view	0.0	93.3	100.0

Table 17: Confirmation set. Net use D is in pp, and permitted use and partial-event retention are percentages. Retention excludes Hear.

Boundary	Method	Inaccessible D	Accessible target use
Hear	Full	57.2 [54.7, 59.6]	89.7 [86.6, 92.1]
Hear	Prose+flag	34.4 [31.2, 37.6]	93.4 [90.7, 95.7]
Hear	Self-filter	47.7 [44.6, 50.9]	93.5 [91.1, 95.7]
Hear	Hard removal	0.1 [0.0, 0.3]	89.1 [85.4, 92.0]
Hear	Character view	0.1 [0.0, 0.3]	89.0 [85.3, 92.0]
Understand	Full	31.0 [27.7, 33.9]	91.5 [88.5, 94.0]
Understand	Prose+flag	32.3 [29.3, 35.1]	95.6 [93.7, 97.3]
Understand	Self-filter	2.0 [0.9, 3.4]	62.9 [59.6, 66.1]
Understand	Hard removal	0.0 [0.0, 0.0]	87.8 [84.1, 91.0]
Understand	Character view	-0.1 [-0.3, 0.0]	87.8 [84.2, 91.0]
See	Full	45.7 [41.5, 49.5]	79.3 [75.2, 83.2]
See	Prose+flag	27.2 [24.6, 29.6]	96.6 [95.2, 97.7]
See	Self-filter	22.2 [19.8, 24.6]	81.5 [78.2, 84.5]
See	Hard removal	0.0 [0.0, 0.0]	80.6 [76.4, 84.4]
See	Character view	-0.1 [-0.3, 0.0]	80.5 [76.4, 84.4]
Receive	Full	56.8 [53.7, 59.6]	94.2 [91.0, 96.6]
Receive	Prose+flag	31.3 [29.2, 33.4]	96.8 [95.6, 97.8]
Receive	Self-filter	4.8 [3.2, 6.6]	53.0 [48.9, 57.2]
Receive	Hard removal	-0.1 [-0.3, 0.0]	90.9 [87.6, 93.7]
Receive	Character view	0.0 [-0.3, 0.3]	90.9 [87.6, 93.7]

Table 18: Five methods on six discovery models and 48 matched scenes (1,152 endpoints per cell). Net use is in pp and permitted use is a percentage. Intervals are 95% paired-bootstrap intervals.

Boundary	Method	Delay 0	Delay 2	Delay 4
Hear	full	69.1 [65.2, 72.7]	68.8 [65.6, 71.9]	71.1 [68.0, 73.8]
Hear	view	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Understand	full	42.6 [36.7, 48.1]	52.7 [48.1, 57.0]	58.2 [53.1, 63.3]
Understand	view	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
See	full	55.5 [49.6, 61.3]	50.0 [44.1, 55.9]	50.8 [43.8, 57.0]
See	view	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Receive	full	68.0 [63.7, 71.9]	68.4 [63.7, 72.7]	69.5 [66.0, 72.7]
Receive	view	-0.4 [-1.2, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]

Table 19: Delayed net use (pp), 32-scene subset, with 95% intervals. The original event remains in context throughout the scripted exchanges.

Boundary	Method	Fixed-A choice contrast	95% CI	Valid choices (%)
Hear	full	63.3	[53.9, 71.1]	99.6
Hear	view	0.8	[0.0, 2.3]	96.9
Understand	full	54.7	[47.7, 60.9]	100.0
Understand	view	0.0	[0.0, 0.0]	100.0
See	full	49.2	[43.0, 54.7]	100.0
See	view	0.0	[0.0, 0.0]	99.2
Receive	full	55.5	[46.9, 64.1]	99.6
Receive	view	0.0	[0.0, 0.0]	97.7

Table 20: Fixed-A decision contrast (pp) and response validity. The choice of location A is tracked across versions, independently of its option number.

Input component	Exact saved text or specified substitution
Role system prefix	`You are role-playing Mira Sen. Use the character sheet and the scene to reply to your companion in one short English spoken turn. Return only the spoken words.`
Shared character	`[Character] Mira Sen is an original fictional volunteer in Harrowbay, a contemporary coastal civic district. They are practical, polite, and concise. They rely on their own observations. Their history and relations are defined only by this scene.`
Full scene (A, no access)	`[Scene state] {"character_id": "Mira Sen", "events": [{"actor": "Alex", "audible_to_Mira Sen": false, "boundary_channel": "Hear", "content": "The glove is in the church.", "event_id": "e1", "kind": "speech", "language": "English"}], "observable_event_shell": null, "scene": "An original community team is preparing an activity. Its members have not shared the latest finding."}`
Role user message	`What information did you get about the item and its location?`
Probe prefix	`Read the scene as an evaluator. Answer the access question with only Yes or No.`
Probe question	`Did Mira Sen hear the utterance?`
Counterfactual B	`The glove is in the church.` → `The broom is in the stairwell.`
Accessible control	`audible_to_Mira Sen: false` → `true`
Inaccessible view scene	`[Scene state] {"character_id": "Mira Sen", "observations": [], "scene": "An original community team is preparing an activity. Its members have not shared the latest finding."}`

Table 21: Exact messages for the selected hearing case, with A/B and access-state substitutions. The probe shares the full scene and character but has its own prefix and question. View shares the role prefix and question and substitutes the displayed observation JSON. Accessible view contains the permitted event in its observation list.

Twin	Access	Probe	Full-context spoken reply	Character-view spoken reply
A	No	No	I heard Alex mention the glove is in the church, though he wasn't speaking to me directly.	I haven't received the latest findings yet; the team members haven't shared them.
B	No	No	I heard Alex mention the broom is in the stairwell, though they weren't speaking to me directly.	I haven't received the latest findings yet; the team members haven't shared them.
A	Yes	Yes	Alex mentioned the glove is in the church.	Alex mentioned the glove is in the church.
B	Yes	Yes	Alex mentioned the broom is in the stairwell.	Alex just told me the broom is in the stairwell.

Table 22: All saved content replies and independent probe answers for scene C001, Qwen3.5-9B, access flag, no delay. Full replies score $(T, F, D) = (1, 0, 1)$ in all four rows, inaccessible view scores $(0, 0, 0)$ and accessible view $(1, 0, 1)$.

Boundary	Method	A first (18 scenes)	A second (14 scenes)	Equal-position effect
Hear	full	65.3 [56.9, 72.2]	60.7 [42.9, 75.0]	63.0 [53.3, 71.2]
Hear	view	0.0 [0.0, 0.0]	1.8 [0.0, 5.4]	0.9 [0.0, 2.7]
Understand	full	56.9 [50.0, 63.9]	51.8 [39.3, 62.5]	54.4 [47.1, 60.9]
Understand	view	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
See	full	50.0 [43.1, 56.9]	48.2 [39.3, 57.1]	49.1 [42.8, 54.9]
See	view	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Receive	full	59.7 [50.0, 69.4]	50.0 [35.7, 62.5]	54.9 [45.5, 63.1]
Receive	view	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]

Table 23: Option-position sensitivity (pp, 95% intervals). A/B order and the question are identical within every pair. The equal-position estimate weights the two order strata equally and resamples scenes independently within each stratum.

(a) Discovery scenes

	Base	Sys.	Sheet	State	Quest.	All	audible
Gemma-4-31B	0	0	1	61	14	100	83–100
Qwen3.8-27B	9	10	68	19	9	78	80–100
Falcon-H1-34B	31	26	70	87	100	100	98–100
Seed-OSS-36B	60	47	85	71	100	95	95–100
Mistral-Small-24B	72	64	73	73	100	100	77–100
Gemma-4-12B	78	88	86	76	100	100	77–100
gpt-oss-20b	80	85	92	82	72	99	92–99
Qwen3.5-9B	85	80	98	90	100	99	95–100
GLM-4.7-Flash	88	86	93	68	100	98	69–100
Nemotron-3-Nano	97	99	97	100	100	99	98–100
Phi-4	98	95	94	86	100	100	93–100

(b) New scenes, characters renamed

Base	Sys.	Sheet	State	Quest.	All	audible
0	0	0	96	8	100	92–100
6	10	72	12	0	71	96–100
25	21	78	98	98	100	100
57	54	74	73	98	90	88–100
81	79	79	90	100	100	98–100
100	100	100	100	100	100	100
100	89	90	99	87	100	94–100
100	100	98	100	100	100	100
97	99	96	91	100	97	95–100
100	100	92	100	100	100	92–100
98	100	96	100	100	100	96–100

Figure 10: Bridge experiment for all 11 open-weight models. Net use of unheard speech (pp) under Study 2's hearing prompt (Base), with one Study 1 feature swapped in, and with all four. Outlined cells differ from Base with a paired 95% skeleton-bootstrap interval that excludes zero. The last column gives the range of target use for the same speech when audible (%). (a) 96 discovery scenes. (b) 96 confirmation scenes with each character renamed to a Study 1 persona.

Model	Base	Sys.	Sheet	State	Quest.	All
Claude Sonnet 5.5	0.5	**9.9**	**15.1**	**8.9**	**9.4**	**13.0**
Gemini 3.1 Pro	0.0	–	0.0	0.0	–	**21.4**

Table 24: Single-feature swaps for two proprietary models on the discovery hearing prompts (192 unheard prompts per cell, net use in pp). Bold marks a change from Base whose paired 95% interval excludes zero. Gemini 3.1 Pro was run on the two swaps that moved the open-weight models most.

Model	Study 1 prompt	+ channel	+ question	+ minimal	All three
Not-audible flag					
Gemma-4-31B	52.1	52.1	51.0	54.7	50.0
Qwen3.8-27B	71.4	74.0	80.2	51.6	32.8
Seed-OSS-36B	90.1	88.0	100.0	87.0	83.3
Phi-4	99.0	100.0	94.3	100.0	94.3
Not-understood flag					
Gemma-4-31B	6.2	3.1	1.6	3.1	0.0
Qwen3.8-27B	35.9	34.4	39.6	12.0	14.1
Seed-OSS-36B	69.8	67.2	95.3	37.0	31.8
Phi-4	88.5	89.6	66.1	69.8	49.5

Table 25: Format ablation from Study 1 toward Study 2 (96 skeletons, content question). Net use (pp) when one Study 2 feature is added (a field naming the channel, the direct question, a minimal context) and with all three.